NORTH CAROLINA RESEARCH

AEROSPACE & AVIATION

GE Aerospace (Durham, NC)
Jet engines (e.g., CFM LEAP, GEnx, CFM56, GE90)

Honeywell (Morrisville, NC – RTP area)
Aerospace technologies and systems (avionics, propulsion & control systems for aircraft)

General Dynamics (Raleigh, NC)
Defense, aerospace engineering and weapons systems support

Boom Supersonic (Greensboro, NC)
Supersonic commercial aircraft development

Collins Aerospace (Charlotte, NC)
Avionics, communication, flight controls for commercial & military

HondaJet (Honda Aircraft Company) (Greensboro, NC)
Light business jet manufacturing and design

ENERGY SYSTEMS

GE Vernova (Wilmington, NC)
Energy technologies (gas turbines, grid solutions, renewable systems)

GE Hitachi Nuclear Energy (Raleigh/Durham)
Nuclear power systems, reactors and advanced nuclear technologies

Duke Energy (Durham, NC)
Electric power and natural gas utility across the Carolinas.

Siemens (Raleigh/Wendell, NC)
Grid systems, power generation, smart infrastructure.

BIOTECH & PHARMACEUTICAL

Amgen (Holly Springs, NC)
Biomanufacturing of innovative medicines

Biogen (RTP / Durham & Wake County, NC)
Biopharmaceuticals focused on neurological diseases

Becton Dickinson (BD) (Morrisville / RTP, NC)
Medical technology & devices (e.g., syringes, diagnostics and laboratory systems)

Eli Lilly (Concord, NC)
Pharmaceuticals (wide range of medicines including endocrinology/diabetes care products and oncology)

Novartis (Durham, NC)
Medicine company specializing in oncology, immunology, and cardiovascular care

Pfizer (Sanford & Durham, NC)
Pharmaceuticals and vaccine manufacturing

ENGINEERING CONSULTING

Alten Technology (Greensboro, NC)
Engineering & technology consulting across aerospace, defense, automotive, medtech, and energy sectors.

Burns & McDonnell (Cary/Raleigh, NC)
Engineering, design, and construction services – power systems, industrial facilities, environmental and infrastructure projects.

HDR (Raleigh, NC)
Engineering, architecture, consulting and project management across infrastructure, energy, water and environment.

Interview Prep Checklist

BD | Senior Engineer - Material Testing | R-538397

- ❑ One Page Graphics
  - ❑ Dewar Production Overview
  - ❑ Material Selection Case Study
  - ❑ Laser Welding Overview

  Response Reps
- ❑ Career Overviews
  - ❑ Manufacturing
  - ❑ Materials & Process
  - ❑ Design
- ❑ Company & Role Specific Questions
  - ❑ Why BD
  - ❑ Why Material Testing Engineer
- ❑ Anticipated Questions
  - ❑ Industry Transition
- ❑ Material & Testing Experience
  - ❑ Examples throughout Career
  - ❑ Non Linear Material & Plastics
  - ❑ Design of Experiment Walkthrough
- ❑ Behavioral Q's
- ❑ Situational Q's
- ❑ Technical Q's
- ❑ Resume Rehearsal
- ❑ Project Arsenal
  - ❑ XDR
  - ❑
  - ❑
  - ❑
- ❑ Questions for them
- ❑ Adjacent Topics

Interview Prep Checklist

BD | Senior Engineer - Material Testing | R-538397

Topic	Context	What Happened	Takeaway
Manufacturing Overview	First Job Out of College Joined a team that needed support for the production of a big space program (JWST)	The official name of the product was called a Integrated Dewar Cooler Assembly which is a fancy term for an infrared sensor and I'm happy to go into the weeds if you want me to But basically it was a huge program for the site and we were only building a few units maybe like 10 in total but my role was to know everything that happened to each unit and make sure it went through the production line smoothly.	And looking back on it, it was honestly one of the best jobs I could've gotten out of college because while there were some stressful moments of being so visible, I was able to learn so much about product development and see a product go through different stages of its build whether its checking the quality of our incoming parts from our suppliers to validate our product through testing, I think my biggest takeaway from that Manufacturing Engr role was getting to see the entire product lifecycle.

PHONE SCREENING GUIDE

BD | Senior Engineer - Material Testing | R-538397
2/27/26 @ 12:30PM PST w/ Jason L

Introduction:

About Me:

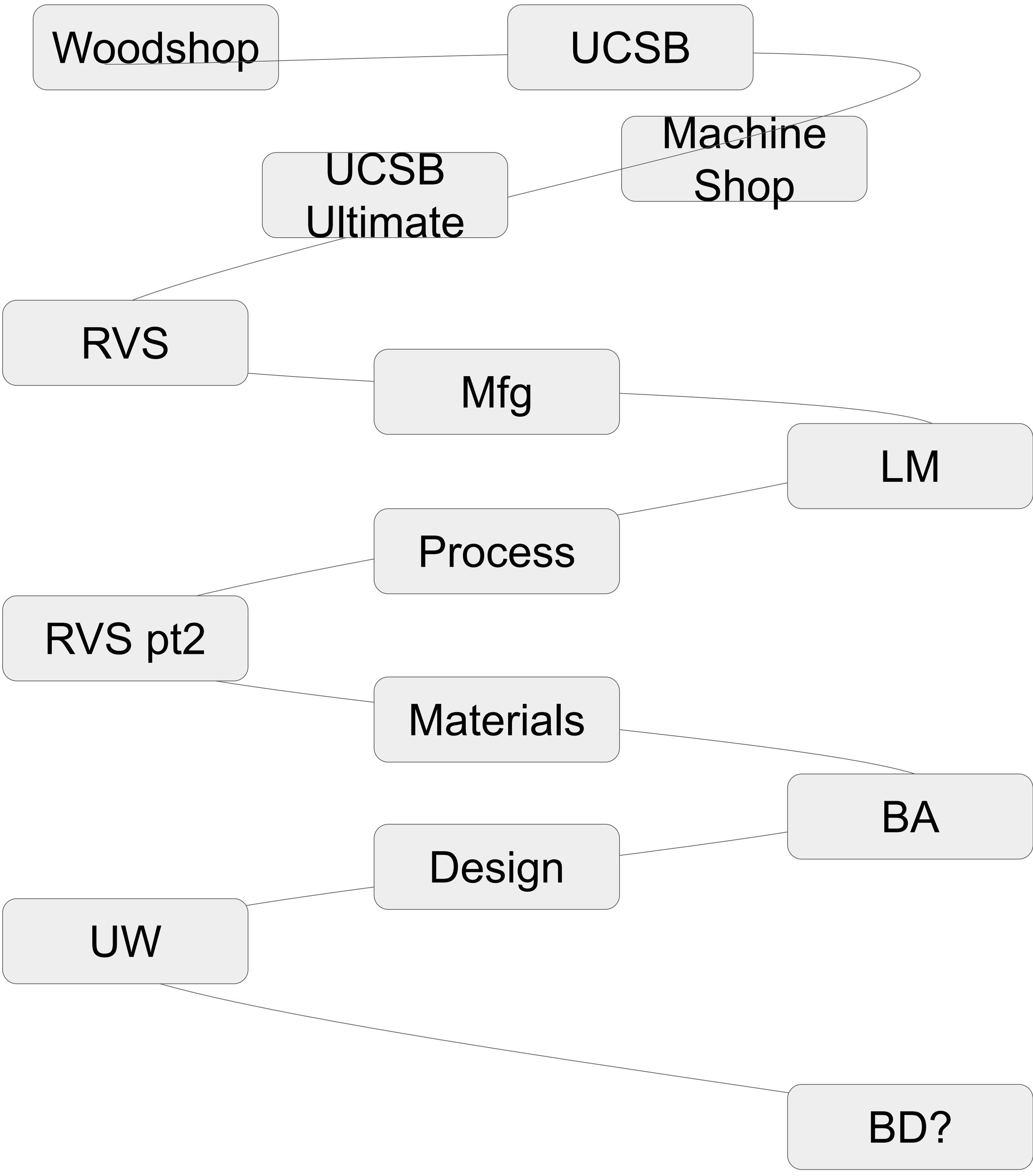
My fit with this role:

Logistical:
- Salary Expectations
- Start Date / Notice Period

Questions:
- Interview Process

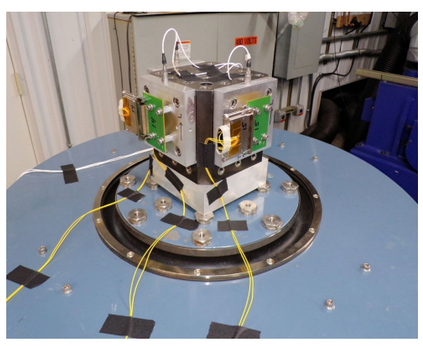
Close Out:

BW Career Map



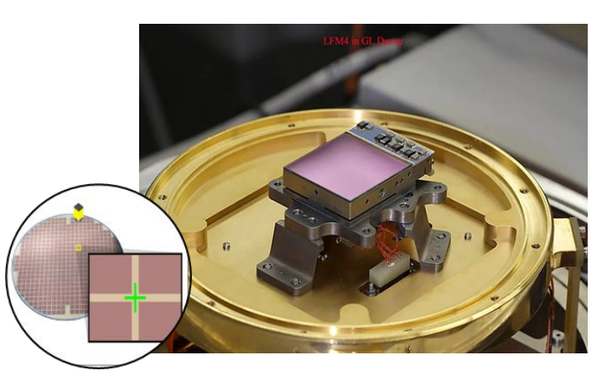
Short Version:
Long Version:

IDDCA PRODUCTION FLOW

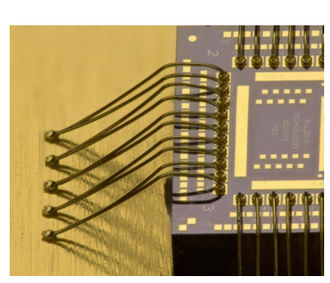


Vibration Testing

Thermal Cycling

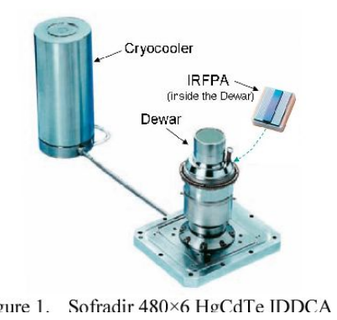


Precision Alignment

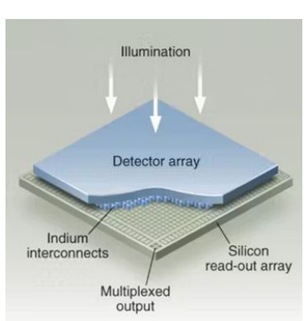


Wirebond

Welding

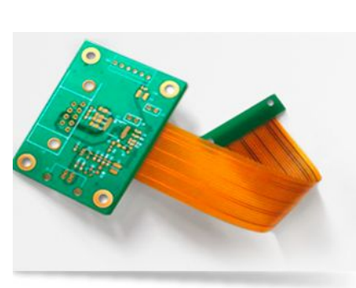


Cryocooler

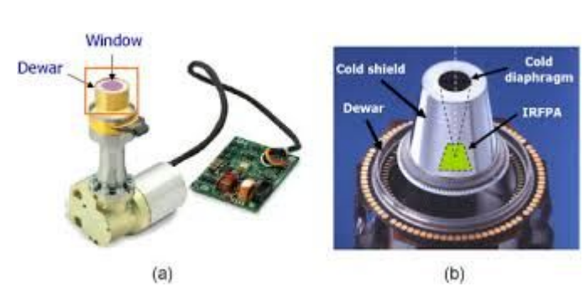


FPA
(ROIC & Detector)

LVA
(Lower Vacuum Assy)



RFA
(Rigid Flex Assembly)

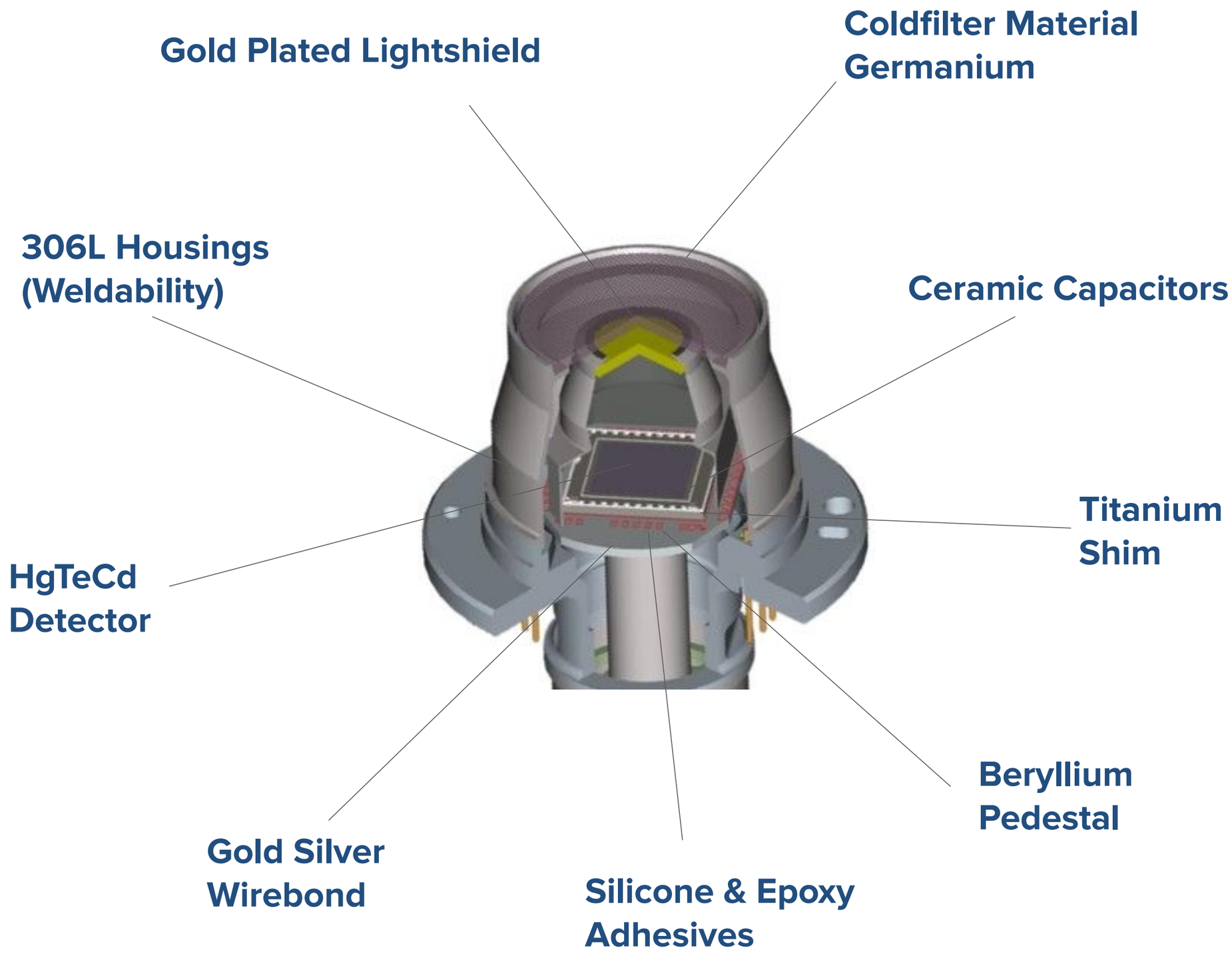


UVA
(Upper Vacuum Assy)

Material Selection Case Study

Integrated Dewar Cooler Assembly

Purpose: Show how much consideration goes into the materials of a product in aerospace. How tradeoffs between material properties drive engineering decisions

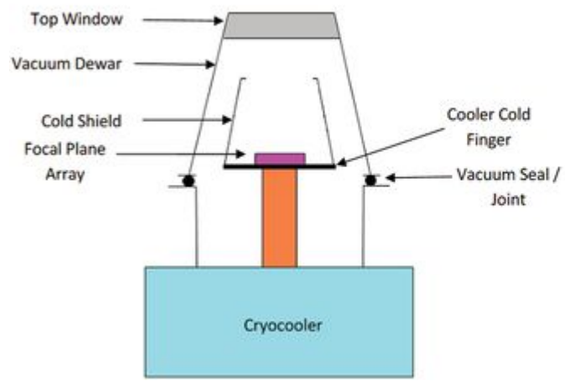


#	Feature	Material	Reasoning & Alternative Differences
1	Detector	HgCdTe	MerCadTelluride bandgap allows for wide range of IR wavelength (SWIR, MWIR, LWIR) & high quantum efficiency... less sensitive to Dark Current
2	Pedestal	Beryllium	Stiffness to Density Ratio
3	Lightshield	Gold Plated	Gold Property of Radiation Shielding
4	Filter	Germanium	Filter Material determines wavelength
5	Housing	Kovar & Alloy 42	Reliable Weldability Properties
6	Shim	Titanium	CTE of Titanium adjust for temperature between Detector & Pedestal
7	Wirebond	Gold Silver	Electrical Conductivity
8	Capacitors	Ceramic	Low Coefficient of Thermal Expansion
9	Adhesive	Silicone	

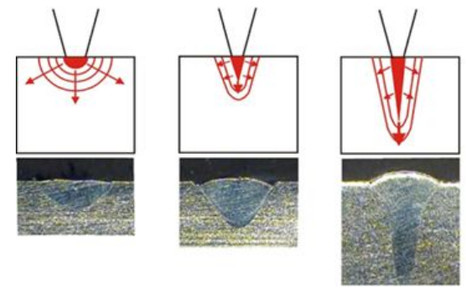
Laser Welding Overview

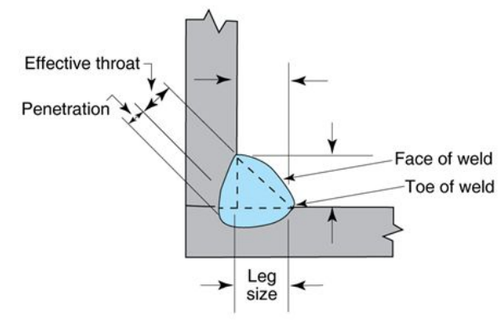
The What

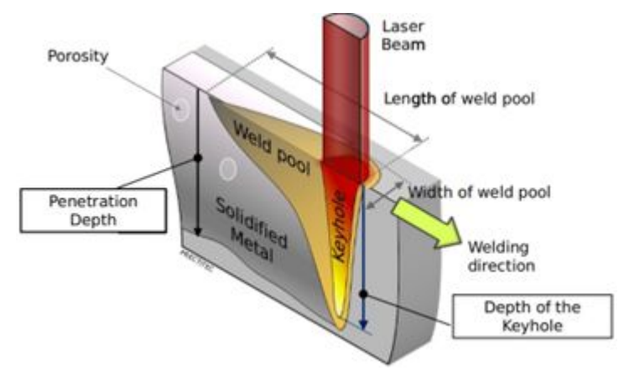
The Why



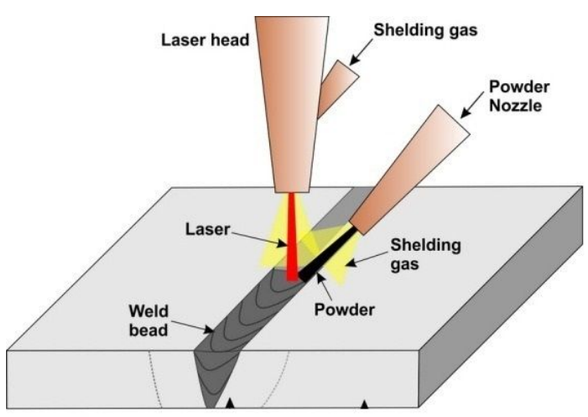
The How

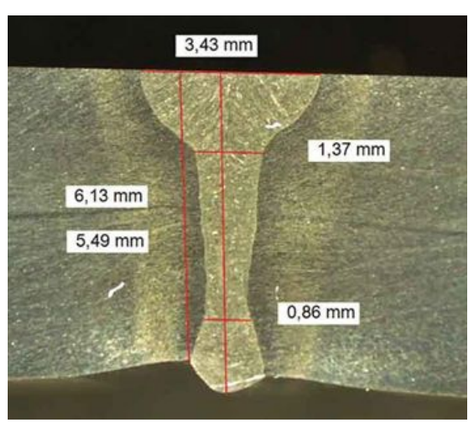






Testing Methods





Material Focus

Material Selection: Design Risk for Initial Weld

	Alloy 42	304L	Kovar	Ti (6Al4V)	Invar 36	Inconel
Alloy 42	Low	Low	Low	High	No Data	Med
304L	Low	Med	Low	No Data	Low	High
Kovar	Low	Low	Low	No Data	No Data	Med
Ti (6Al4V)	High	Not Data	No Data	Low	No Data	No Data
Invar 36	No Data	Low	No Data	No Data	No Data	No Data
Inconel	Med	High	Med	No Data	No Data	High

Standards

AWS D17.1/D17.1M
Specification for Fusion Welding for Aerospace Applications

ISO 15614-11
Specification and qualification of welding procedures for metallic materials — Laser beam welding

AS9100: Quality Management Systems – Requirements for Aviation, Space, and Defense Organizations

ISO9001: the International Standard for Quality Management Systems

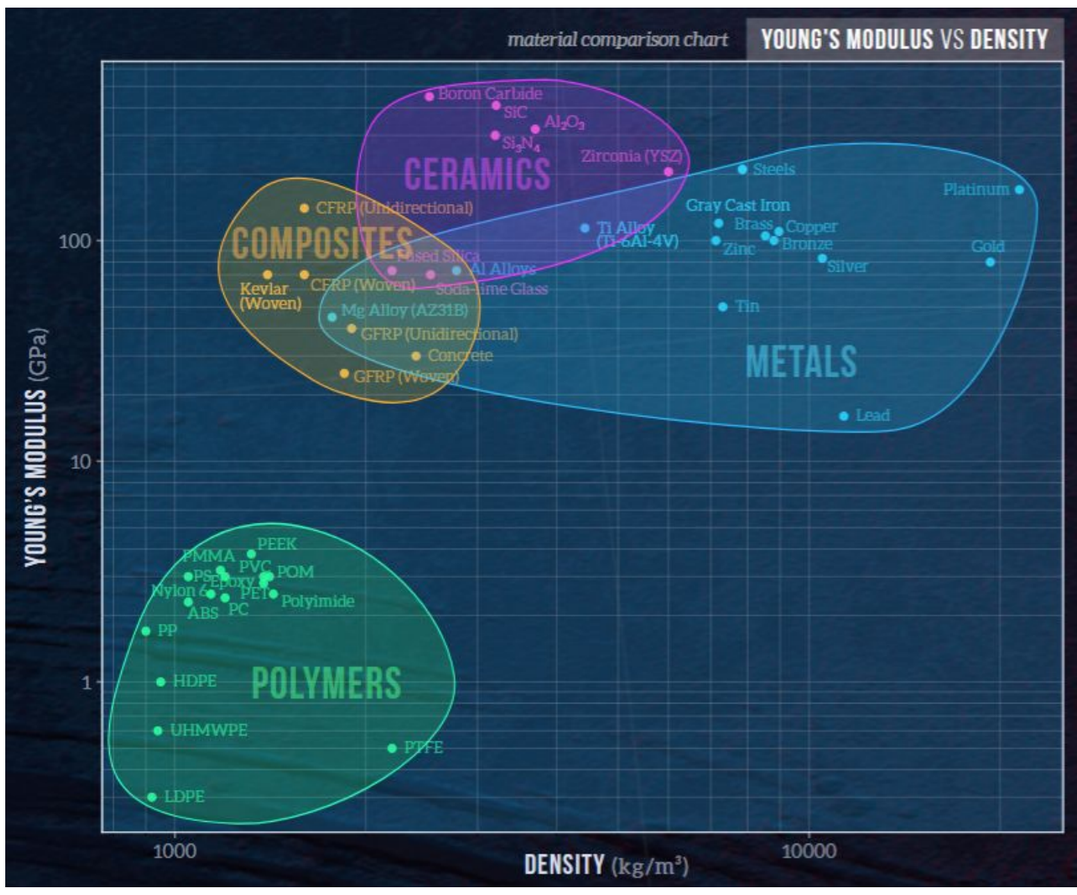
ASME 3.3.1

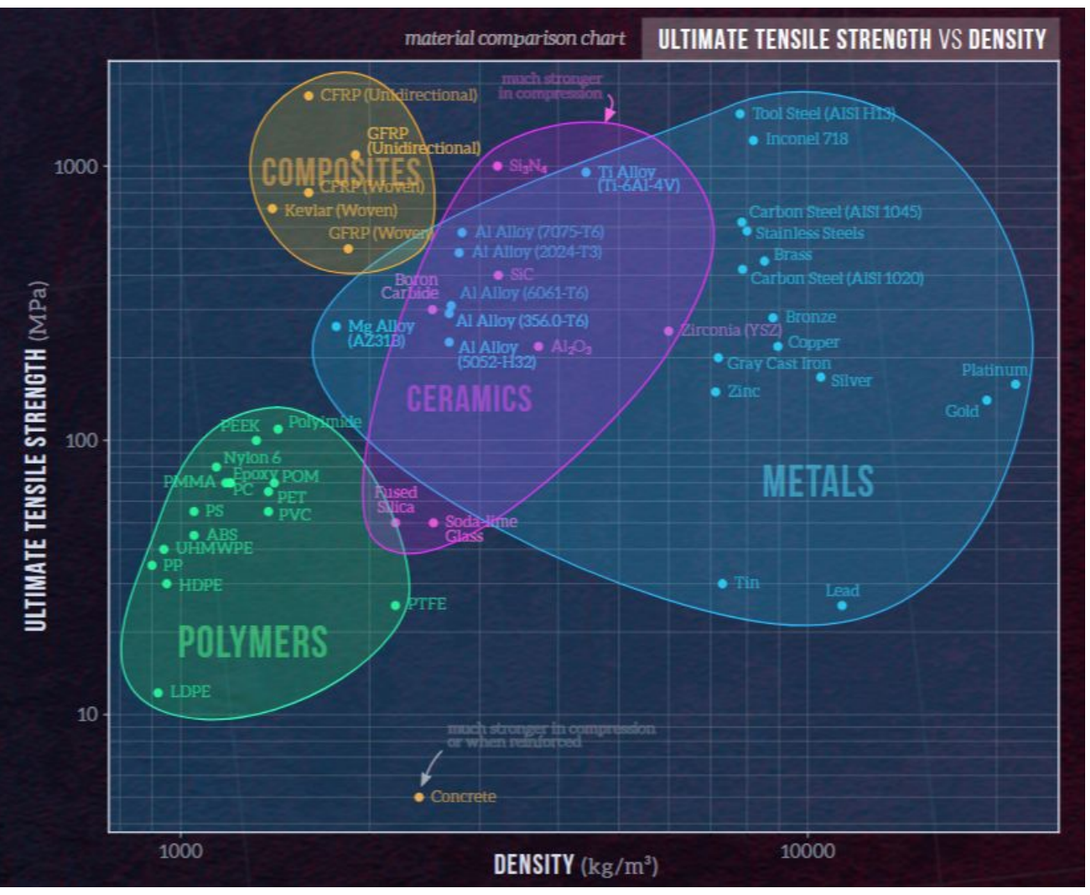
ANSI/NCSL Z540-1 General Requirements for Calibration Laboratories and Measuring and Test Equipment

Takeaways & XRD

Polymers vs Metals

Purpose: Show how Plastics and Metals actually have a lot in common

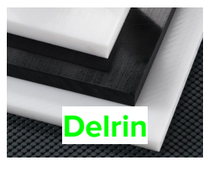




Polymers

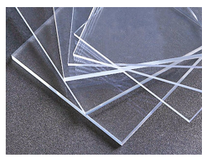
Polysulfone (PEEK)

Acetal (POM)



PTFE (Teflon)

PMMA (Acrylic)



Polycarbonate (PC)

ABS

Polyimide

Epoxy

Metals

Crystallinity: Higher Crystallinity = stronger/stiffer/higher melting pt	Structure = Strength	Grain Structure: Smaller Grains = Stronger (Hall-Petch)
Cold Drawing during necking	Stronger using Plastic Deformation	Cold Working to increase strength (Rolling & Forging)
LDPE & HDPE have same chemical composition but different crystanility changes strength, stiffness, hardness, transparency	Similar Chemical Composition but different internal structures	Steel + Heat Treatment create different microstructures (Ferrite, Pearlite, Martensite) Changes strength, toughness, hardness, ductility
Vulcanization: Heating Rubber in presence of sulfur to simulate cross-links and add stiffness in a rubber	Restricting Internal Movement Increases Strength	Precipitation Strengthening: Heat Treatment process uses dissolved impurities to improve mechanical properties of metal alloys
Ductile to Brittle Transition Temperature (DBTT)	Temperature Points	Glass Transition Temperature (T_g)

Takeaways

Explains why Heat Treatment matters

Explains why Processing Techniques matter

Explains how Identical Ingredients do not guarantee identical performance & material properties

BW Career Overview

- Manufacturing
  - Biggest Takeaway: Learning a Product Lifecycle & Taking a Product from Craddle to Grave
    - Explaining how a FPA Dewar is manufactured
      - Focus on Testing Areas & Coupons
  - Resume:
    - EVMS Budgeting / Cleanroom / Prototype Builds / Suppliers
    - Typical Mfg Practices: Lean Practices & RCCA & DFMA

- Materials & Process
  - Biggest Takeaway: Diving deep into a process & taking ownership of a specific part of a product build
    - Laser Welding
      - Material Science
      - XRD
    - Precision Alignment
      - Data & Tolerancing
      - Matlab
      - Automation
  - Resume:
    - Vibration / Thermal Cycling / PRR/CDR/TTRs / SEM & Failure Analysis / Automation / Capital Equipment / SPC & Cpk / Custom Test Fixtures / Material Characterization & Metallurgical Analysis / V&V Frameworks

- Tooling Design
  - Biggest Takeaway: Learning a new product and the world of design - No right path to the same answer // Tradeoffs
    - Material Selection Decisions
  - Resume:
    - 3D Modeling / FEA Stress Analysis / Fatigue Analysis

BW Career Overview

Materials

Testing

Manufacturing

- Material Selection within Dewar Product
-

- Product Lifecycle Testing
- Critical Gate Coupons
- Non Recurring Tests

Materials & Process

- Welding Material Compatibility

- Laser Welding Process Sustaining
- XRD

Design

- Material Selection in Design
- Composites
- 3D Printing Plastics

- FEA & Stress Analysis
- Fatigue Testing

BD General

BD Specific

Why BD?

- Relocation with NK
- Transitioning to a new industry
- Aligned Company Mission
  - Pride in MedTech Products
  - High Stakes Environment & scrutiny of medical devices
- Heard great things about BD Culture from Anita & NK

What excites you about this role specifically?

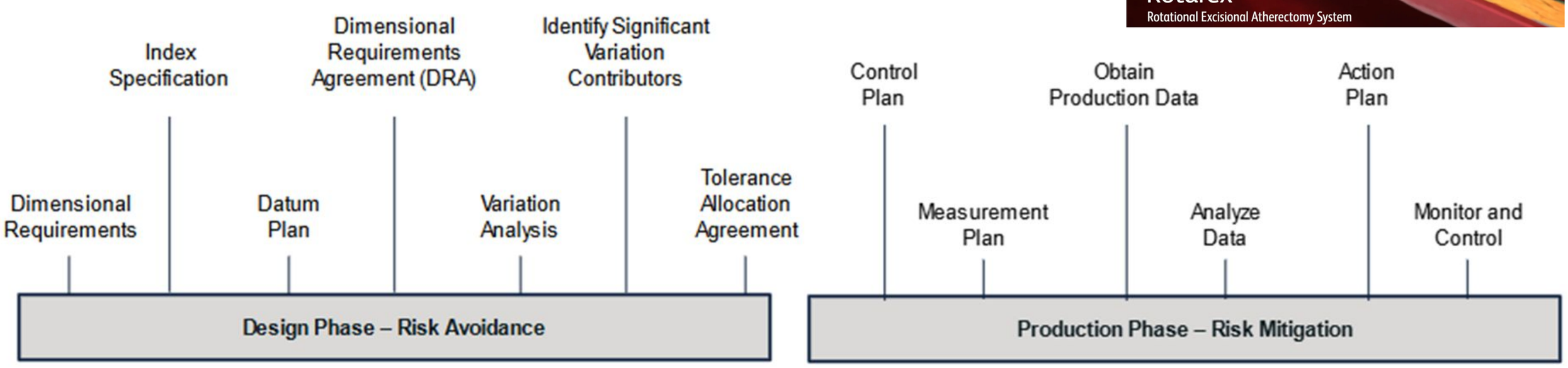
- Aligns with various interests
  - Balance of being Hands On & Design/Analysis
  -
- Growth
  - New Industry
  - New Materials
  - within DE S&T Group (Material Testing & Simulation)

Products that excited me:

Rotarex™ Rotational Excisional Atherectomy System

A product that removes plaque buildup from clogged arteries, re peripheral artery disease (PAD) or coronary artery disease

Venous Technology



BD Anticipated

BD Specific

Why BD?

- Relocation with NK
- Transitioning to a new industry
- Aligned Company Mission
  - Pride in MedTech Products
  - High stakes environment
- Heard great things about BD Culture from Anita & NK

What excites you about this role specifically?

- Aligns with various interests
  - Balance of being Hands On & Design/Analysis
- Growth within DE S&T Group (Material Testing & Simulation)
  - Working with new materials & new industry

Industry Transition - How would I transition into the MedTech Industry?

- Do what I did whenever I transitioned to a new industry:
- Dedicate the first couple of months to learning and absorbing
  - Learning the Requirements, Processes, Challenges
  - Asking the whys* from different departments & documenting the patterns
  - Building a network of allies and understanding everyone's perspectives & functions
    - I may not know the answer but I might know someone who can point me in the right direction
  - Building my tools - command media, standards, frequently referenced resources, understand what our team is responsible for
- Like Anita mentioned, there is a lot of overlap & parallels between Aerospace & MedTech

Experience with Material Testing?

- See Slide about Material & Testing Chart

What do you know about our team's work?
What is your experience with ASTM or ISO Testing Standards?
Walk me through designing an experiment

BW Behavorial Q's

Behavorial Questions

(teamwork, communication, problem-solving)

Tell me about yourself.

Describe a challenging engineering problem you worked on.

Tell me about a time you failed or made a mistake. What did you learn?

Describe a time you worked on a team with conflicting opinions.

How do you handle tight deadlines or multiple projects?

Give an example of when you had to explain a technical concept to a non-technical audience.

Tell me about a time you took initiative or led a project.

How do you handle feedback or criticism?

What made you switch jobs so frequently? Why haven't you been in a role for more than 2 years?

BW Situational Q's

Situational Questions

(logic, ethics, communication, and practicality) "what would you do if..." questions to test judgment and decision-making

What would you do if your design didn't meet safety requirements late in the project?

How would you handle discovering an error in your calculations after submission?

If a teammate isn't pulling their weight, how would you handle it?

How would you prioritize tasks when multiple projects are due at the same time?

What would you do if manufacturing tells you your design is too expensive to produce?

How would you respond if your manager disagreed with your technical recommendation?

How do you approach designing an experimental plan for an ambiguous or poorly defined problem?

BW Technical Q's

Technical Q's

(

What is your experience with ASTM or ISO Testing Standards?

Engineering Cheat Sheet & Classic Mechanical Concepts & Examples

BW Resume

Tooling Design	
3D Modeling	
Tolerance Stack Up	
DFM Principles	
Design Intent & Physical outcomes	
FEA Ansys (static/dynamic/thermal loading conditions)	
Stress Analysis	
Fatigue Analysis	
PLM (Product Lifecycle Management)	
Process Engineering	
Metallurgical Analysis	
Thermal Analysis	
Hardness Testing	
Microstructure Evaluation	
Custom Test Fixtures	
SPC & Cpk Analysis	
DOE-based characterization workflows	
Capital Equipment Project	
Semiconductor	
Automated Assembly Lines	
Scanning Electron Microscopy (SEM)	
PRR/CDR/TRR Reviews	
V&V Frameworks	
Manufacturing	
EVMS & Budgets	
Class 7/Grade C/ Lvl 3 Cleanroom	
Additive Manufacturing	

BW Project Arsenal

Project Category: Technical Win, Leadership,

Laser Weld Capital Equipment Project
Situation / Context
Context: Two Laser Welding machines - 181 & IPG
181 was on life support. Noticeably having more maintenance issues with lightbulb, obsolete controllers
After Ultra I incident, Leadership wanted to identify aging equipment and potential single point failure
181 identified as one of those machines
Task / Gameplan
Key Decision: Procure a new Laser Welder
Providing Justification, Working with the Vendor, Coordinating with Factory & Facilities, Notifying affected programs
Presenting to PCCB for capital investment justification
Action / Running the Play
Coordinating with Vendor on Site Acceptance Testing
Hiccup: Power settings were not directly translatable 1:1 due to different quality of fiber beams ➔ Solution: Run several coupons to achieve similar penetration depth and understand power settings & cross section
Tedious Tasks: Each Program had to have two separate VDD despite our initial intent of having flexibility of using the same welder model
Result / Takeaway
Lesson: What I thought was going to be a relatively simple process of copy and paste turned to be one of the most extensive projects mainly because there were so many factors I didn't expect. But I realized buying a capital equipment isn't just a machine but an investment for that site.
That's It

BW Project Arsenal

Project Category: Trade Off & Cross Department Conflict

Smart Scopes vs Legacy Ultras
Situation / Context
Process: Precision Alignment - We had two main portfolio of products: Tactical: High Rate, High Volume Space: High Cost, Low Volume
There was one specific Cost Plus Space program that drove the development of future follow on fixed firm price programs. Had their own Alignment Z-Scopes Started to become an issue with calibration and property ownership Meanwhile, Tactical Programs had adapted to a more modern scope QVI
Task / Gameplan
So Alignment Team were at crossroads for the new Space Programs: Adopt a new modern scope like the QVI or continue working with the legacy scopes with the risk of calibration times and ownership issues
Key Decision: Transfer Programs to QVI Smart Scopes and prove capability in parallel with using the Legacy Smart Scopes
Action / Running the Play
Obtain Mechanical Samples and Develop Programs for the new QVI Scopes
Perform Gage R&R studies & Compare to Legacy Scopes
Measurement System Analysis - Uncertainty with Tolerances
Result / Takeaway
Process Control Change Board - PCCB
“If it ain’t broke, don’t fix it” mentality with different teams
Cross Functional Disagreement - all want the same goal but different perspectives on the path there
Lesson: **That’s It**

BW Project Arsenal

Project Category: Design of Experiment Walkthrough

Laser Weld Z Height Shrinkage

Situation / Context

Process: Laser Welding
There was a question that was asked to Process Engineering about how much Laser Welding impacted the Z-Height Shrinkage and we didn't have the date to back that up it was negligible

Design of Experiment

1) **Establish the question:**
   What is the magnitude of Z-height shrinkage caused by the welding process?
2) Define Variables:

Weld Input & Speed

Material & Thickness

Operator

Response Variable: Z- Height Shrinkage

3) Define Experiment Process

Measure Pre-Weld Height Feature

Perform Weld

Measure Post-Weld Height Feature

Measure across multiple points

Record Variables

Use repeatable measurement system (CMM or Laser Scanner)

Repeat for number of samples and different day

Analyze Results / Takeaway

4) Found a shrinkage of 0.001"
Document Process Notebook
5) Incorporate into Bi-Annual Laser Weld Validation Process

Lesson:
That's It

BW Project Arsenal

Project Category: Technical Win

X-Ray Diffraction (XRD) Project	
Situation / Context	
Task / Gameplan	
Key Decision	
Action / Running the Play	
Result / Takeaway	

Lesson:
That's It

BW Project Repository

Project Repository

(Repository of significant projects & stories that have shaped my career & lessons learned)

Role Specific (Material Testing Engineer):

- **Laser Weld XFD**

Repository:

- **Laser Weld Capital Project**
- **BL0 Project**
- **Working with V**

Secondary Adjacent Topics:

MATLAB /

LABVIEW - Been a while but similar data acquisiton methods in Thermal Cycling

Vibration Testing

Thermal Cycling

Non Linear & Plastics Material Experience

Residual Stress

Metal Materials

BW Questions for them

What are my questions to them?

What are currently the biggest challenges of the team?
(Understanding why the req is open in the first place - is there a big project coming up, is the team understaffed, etc.)

What is the team size like? When was the last time you all celebrated something?
(Understanding team culture & team dynamic - are these people I can see myself working 30% of my life with?)

I've heard one of the challenges of BD is having requirements not just with the FDA but with regulatory bodies around the world, I'm curious if this is an issue with BD and if that's something this team deals with as well?

Claude Generated:
Can you describe how the material testing team interfaces day to day with the simulation engineers - is it a collaborative back and forth throughout the project or more of a hand-off model?

What does the current lab capability look like and where are there still gaps the team is looking to fill?

What would success look like in this role at 6 months and 12 months?

What are the next steps in the process?

BW Strengths & Weaknesses

Strengths

- Driven: Once I have an idea, I become determined, almost obsessive about getting that project done
  - Not afraid to ask dumb questions
  - Not afraid to fail (in the right context) - where we learn the most & gain the most experience
  - Not afraid of new environments or being uncomfortable
  - Not Reinventing the wheel: Reaching out to people who have done something similar
  - Breaking Big Problems into Smaller Problems and back tracking
  - Writing Down my goals and understanding the bigger picture
  - Right is better than rushed

Weaknesses

- Figuring out when Done is better than Perfect
- Sometimes that obsessiveness can go down a deep dive that might not be value added
- Answer what the Customer is asking for
- How I've gotten better:
  - Establishing Must Haves, Nice to Haves, Must Not Haves helps me clearly define the requirements
  - Setting Mini Deadlines for myself
- Controlling what I can control

Flash Cards

- Generic Flashcards
- Resume Flashcards
- Behavioral & Situational
- Technical Engineering Q's